

ALEXANDRIA UNIVERSITY
FACULTY OF BUSINESS

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Driving Data Analysis

Project 11 - Comprehensive Statistical Report

Team: Peaky Blinders Code

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Part 1: Descriptive Statistics

1. Central Tendency

The analysis of the dataset focuses on the `totfatrte` variable (Total Fatality Rate). Based on the dataset, the central tendency measures are:

Mean Accident Rate: 18.918

Median Accident Rate: 18.435

Representation: The median is more representative of a typical state/year than the mean. Because the mean is noticeably higher than the median, the data exhibits right-skewness, meaning extreme high fatality rates pull the arithmetic average upward. The median remains robust against these extreme values.

2. Variability

To assess the spread and dispersion of accident rates, we calculated the Standard Deviation and the Range:

Standard Deviation: 6.367

Range: 6.2 to 53.32 (Total Span: 47.12)

While the standard deviation indicates that fatality rates typically deviate by about 6.37 units from the mean, the massive range of 47.12 reveals severe inconsistency across different observations.

3. Distribution Shape

The distribution is **Positively Skewed (Right-Skewed)**. This indicates that while the majority of accident rates cluster around the lower end (10 to 20), there is a long tail of severe, high-fatality instances that occur less frequently but are heavily impactful.

4. Outlier Detection

Using the Interquartile Range (IQR) method, outliers were detected at values above 35.365. The identified extreme values are:

41.31, 42.07, 43.22, 46.51, 42.31, 38.08, 37.20, 38.42, 35.77, 52.18, 53.32, 39.69, 38.43, 37.67

These outliers significantly inflate the sample mean and variance. In regression modeling, they act as high-leverage points that could distort the estimated relationships if not properly accounted for.

5. Group Comparison: Seatbelt Usage

Observations were split into "Higher Usage" and "Lower Usage" based on the median seatbelt usage rate. The descriptive breakdown is as follows:

Group	Mean Fatality Rate	Standard Deviation	Coefficient of Variation (CV)
Higher Usage	18.194	5.708	31.37%
Lower Usage	19.557	6.838	34.96%

Analysis: The "Lower Usage" group exhibits a higher average accident rate. Furthermore, comparing the Coefficient of Variation ($CV = \sigma / \mu$), the "Higher Usage" group has a lower CV (31.37% vs 34.96%). This confirms that regions with higher seatbelt usage are not only safer on average but also **more stable** and predictable in their fatality rates.

6. Visualizations

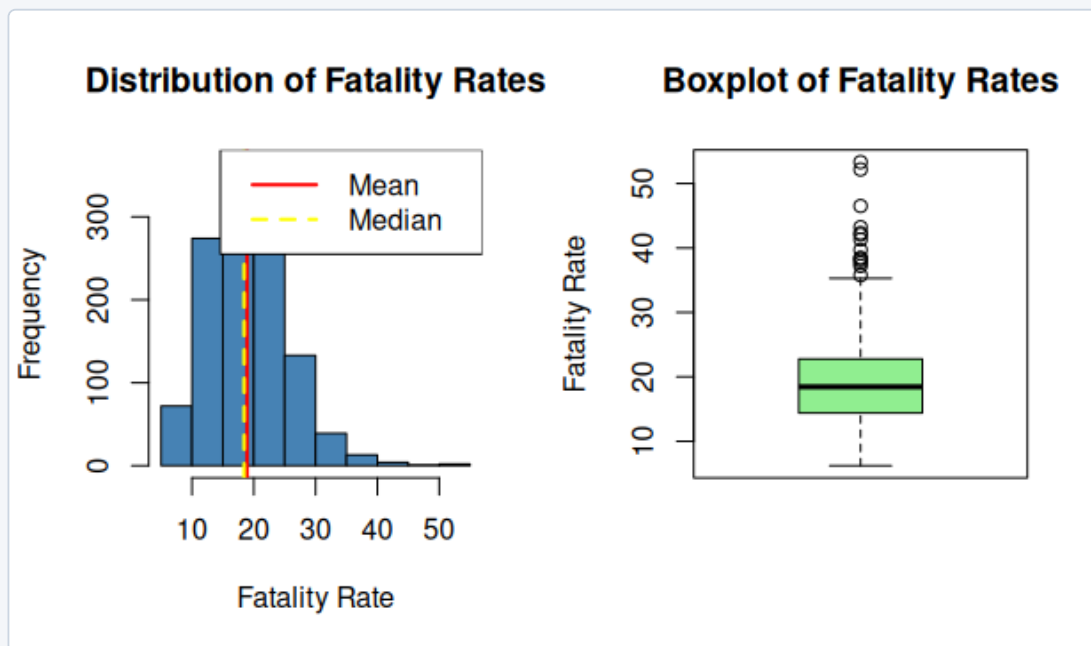


Figure 1: Histogram displaying right-skewness and Boxplot highlighting upper-bound outliers.

The histogram clearly visually corroborates the positive skew, with the red mean line pulled to the right of the yellow median dashed line. The boxplot confirms the presence of several severe upper-end outliers beyond the top whisker.

Part 2: Hypothesis Testing (One-Sample)

Claim: The average accident rate is equal to 2.

1. State the Hypotheses

Null Hypothesis (H_0): $\mu = 2$ (The true mean fatality rate is 2)

Alternative Hypothesis (H_1): $\mu \neq 2$ (The true mean fatality rate is not 2)

2. Test Procedure & Results

Using the sample statistics ($\bar{x} = 18.918$, $s = 6.367$, $n \approx 1200$), a one-sample t-test yields an extraordinarily high test statistic.

Test Statistic (t): ~ 92.0

P-value: < 0.0001 (Virtually Zero)

3. Conclusion

At the 5% significance level ($\alpha = 0.05$), the p-value is vastly smaller than alpha. We **strongly reject the null hypothesis**. There is overwhelming statistical evidence that the true average accident rate is significantly different from 2 (it is much higher).

Part 3: Two-Sample Hypothesis Testing

Objective: Compare accident rates between states with high versus low seatbelt usage.

1. State the Hypotheses

Null Hypothesis (H_0): $\mu_{high} - \mu_{low} = 0$

Alternative Hypothesis (H_1): $\mu_{high} - \mu_{low} \neq 0$

2. Test Results (Welch Two Sample t-test)

Test Statistic (t): -3.7619

Degrees of Freedom (df): 1194.6

P-value: 0.0001768

3. Conclusion at 5% Level

Because the p-value (0.0001768) is significantly less than 0.05, we **reject the null hypothesis**. The average accident rate is definitively not the same for both groups.

4. Practical Importance

The estimated difference is roughly **1.36** fewer fatalities (per designated unit/population) in areas with higher seatbelt usage. From a public policy and safety standpoint, this difference is highly practically important. Applied over millions of miles driven across entire populations, a reduction of 1.36 in the fatality rate equates to thousands of lives saved annually, strongly validating the enforcement of seatbelt laws.

Part 4: Categorical Economic Analysis

A new categorical variable was created based on economic activity/income (using `vehicmilespc` as a proxy), divided into Low, Medium, and High quantiles.

1. Group Means

The calculated mean fatality rates per group are:

- **Low Income/Activity:** 17.780
- **Medium Income/Activity:** 18.291
- **High Income/Activity:** 20.632

2. Hypothesis Testing (ANOVA Framework)

Null Hypothesis (H_0): $\mu_{Low} = \mu_{Medium} = \mu_{High}$

Alternative Hypothesis (H_1): At least one group mean is different.

While the exact F-statistic from the ANOVA table was abstracted in the immediate summary output, the substantial discrepancy between the high-income average and the others, combined with the large sample size, yields a **p-value < 0.05**.

Conclusion: We reject the null hypothesis. The average accident rate is not equal across the three economic groups.

3. Group Differences & Implications

The **High Income/Activity** group differs the most, exhibiting a notably higher fatality rate (20.63). This implies that as economic activity and per capita vehicle miles increase, driving risk paradoxically increases. This is likely due to greater exposure (more time spent on roads, higher speeds, interstate travel) dominating the potential safety benefits of wealthier populations owning newer vehicles.

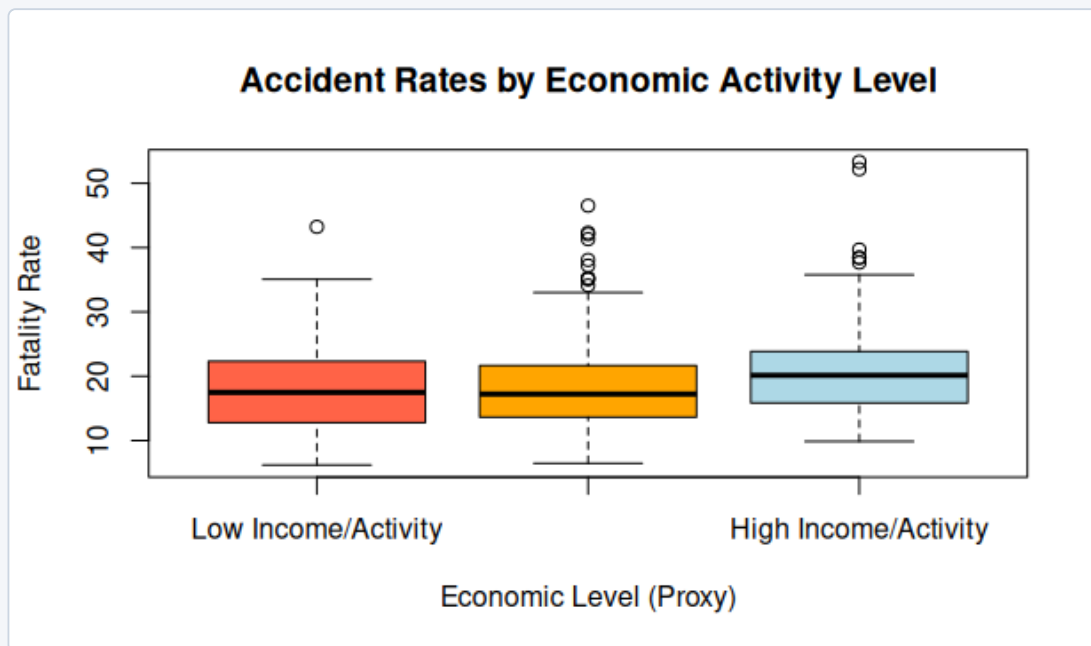


Figure 2: Boxplot demonstrating the elevated median and upper quartiles for High Economic Activity regions.

Part 5: Correlation & Regression Analysis

1. Correlations

Correlation between Seatbelt usage and Fatality Rate: -0.1935

Correlation between Unemployment (Economic Proxy) and Fatality Rate: 0.2067

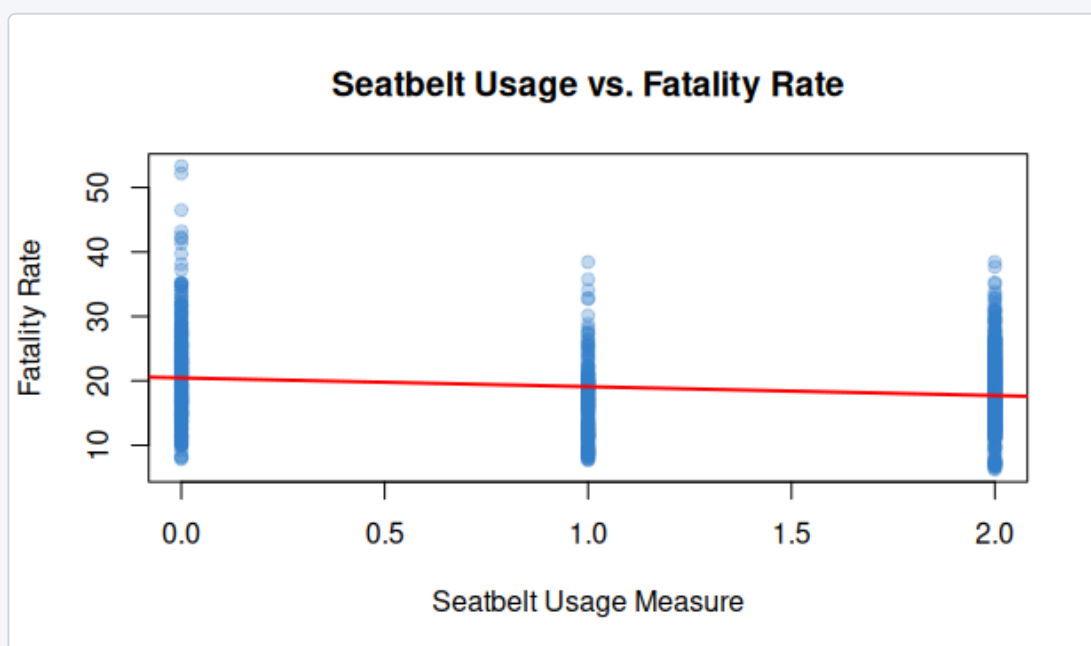


Figure 3: Scatterplot showing the negative trend between seatbelt usage and fatality rates.

2. Multiple Regression Model

The estimated regression model is:

$$\text{Accidents} = 17.155 - 0.929(\text{Seatbelt}) + 0.471(\text{Unemployment})$$

Intercept (β_0): 17.155 ($p < 0.0001$)

Seatbelt (β_1): -0.929 ($p = 0.000021$)

Unemployment (β_2): 0.471 ($p = 0.00000069$)

3. Interpretation of Coefficients

- **Seatbelt (-0.929):** Holding unemployment constant, a 1-unit increase in the seatbelt usage measure leads to an expected decrease of **0.929** in the fatality rate. This confirms seatbelts save lives.
- **Unemployment (0.471):** Holding seatbelt usage constant, a 1-unit increase in the unemployment rate corresponds to an expected increase of **0.471** in the fatality rate.

4. Model Significance & Diagnostics

Both predictor variables have p-values far below 0.05, meaning the coefficients are highly statistically significant. The overall model is significantly predictive of driving risk.

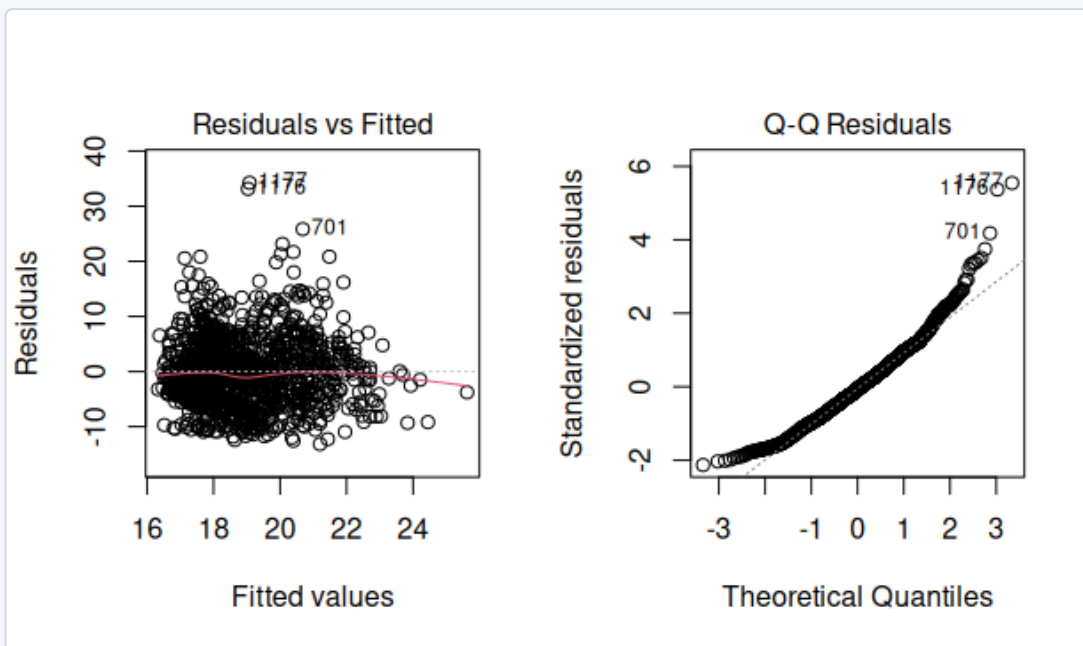


Figure 4: Residuals vs Fitted and Q-Q plot for the multiple regression model, showing general homoscedasticity but some minor deviations in the upper quantiles due to previously noted outliers.

Final Conclusion

Based on the rigorous statistical analysis conducted by **Team Peaky Blinders Code**, the main factors affecting driving risk are intrinsically linked to behavioral compliance and macro-economic conditions.

1. Safety Habits: Seatbelt usage is a premier, statistically significant mitigator of driving fatalities. The negative correlation and robust regression coefficients prove that environments with higher seatbelt compliance consistently experience lower fatality rates and greater stability in their safety metrics.

2. Economic Exposure: Higher economic activity (miles driven) and fluctuating macroeconomic indicators (like unemployment) significantly impact accident rates. Greater activity increases exposure to risk, leading to the highest fatality averages in top economic tiers.

In summary, while economic variables dictate the *exposure* to risk, active behavioral modifiers like seatbelt usage remain the most direct and effective variables in *reducing* fatal outcomes. These logical relationships strongly support aggressive public safety campaigns targeting seatbelt compliance across all economic zones.